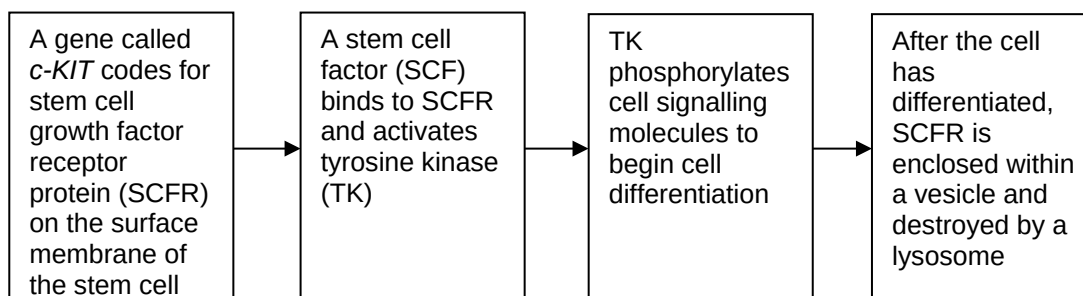


0 4

Following a body injury, bone marrow stem cells move to the site of damage and undergo cell differentiation.

Figure 5 shows how this differentiation occurs.

Figure 5



0 4 . 1

Suggest how SCFR is destroyed by a lysosome.

[2 marks]

After a heart attack, cardiomyocytes (cardiac muscle cells) die, and become infarcted tissue. Infarcted tissue cannot contract.

Stem cells in bone marrow **cannot** move to the infarcted tissue and differentiate into cardiomyocytes.

Scientists used laboratory rats to investigate if bone marrow stem cell transplants could be used to repair infarcted tissue resulting from a heart attack.

They split the rats into three groups.

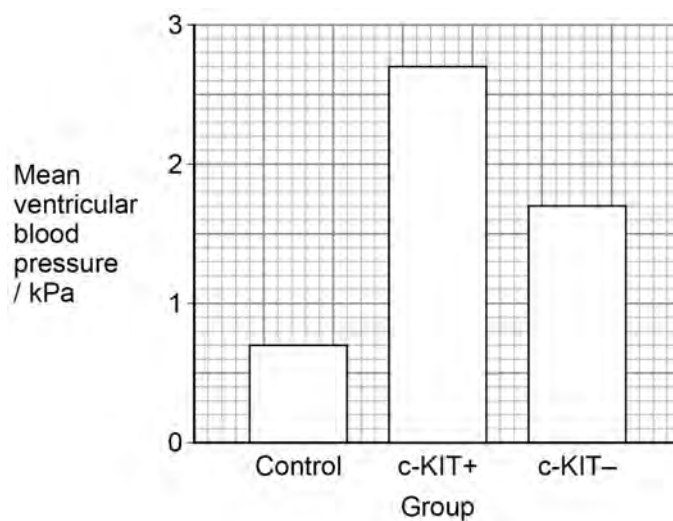
- **Control** group did not get a transplant of bone marrow stem cells.
- **c-KIT+** group got a transplant of bone marrow stem cells with a functioning *c-KIT* gene.
- **c-KIT-** group got a transplant of bone marrow stem cells with **no** functioning *c-KIT* gene.

After 9 days, the scientists measured the mean ventricular blood pressure of each of the three groups.

Figure 6 shows their results.

The differences between the groups were all statistically significant.

Figure 6



0 4 . 2

Using all of the information, suggest explanations for the results for the **Control** group and the **c-KIT-** group shown in **Figure 6**.

[4 marks]

Control

c-KIT-

Turn over ►

0 4 . 3

Nine days after transplantation, the **c-KIT+** group showed that 68% of infarcted tissue was made up of new cardiomyocytes. The control group had **no** new cardiomyocytes.

Assuming that mean ventricular blood pressure is directly proportional to the number of cardiomyocytes, calculate the percentage of infarcted tissue that was made up of new cardiomyocytes in the **c-KIT-** group.

[2 marks]

Answer _____ %

All new cardiomyocytes produced:

- Connexin-43, a channel protein that allows electrical impulses to pass between cardiomyocytes
- GATA-4, a transcriptional factor that stimulates the expression of genes for actin and myosin.

0 4 . 4

Suggest how production of Connexin-43 and GATA-4 could give the result seen in the **c-KIT+** group in **Figure 6** (on page 13).

Do **not** include details of transcription or translation in your answer.

[2 marks]

Connexin-43 _____

GATA-4 _____

Turn over for the next question

*Do n
outs*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►

